JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SATURDAY 04th APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

CHEMISTRY — TEST PAPER WITH SOLUTION

SECTION-A

51. Number of moles and number of molecules in 1.4187 L of SO_2 at STP respectively are

(1) 0.1266; 3.812×10^{22}

(2) 0.0633; 3.812×10^{22}

(3) 0.1266; 7.6238×10^{22}

(4) 0.0633; 7.6238×10^{22}

Sol. $\text{moles} = \dfrac{\text{V lit}}{22.7 \text{ lit}} \Rightarrow \dfrac{1.4187 \text{ lit}}{22.7 \text{ lit}} \Rightarrow 0.0633$

no. of molecules = moles $\times 6.023 \times 10^{23}$

$\Rightarrow 0.0633 \times 6.032 \times 10^{3}$

$= 3.812 \times 10^{22}$

52. What is the ratio of wave number of first line (lowest energy line) of Balmer series of H atomic spectrum to first line of its Brackett series?

(1) 5:1

(2) 5:0.81

(3) 5:1.75

(4) 5:27

Sol. $\bar{V}_1 = R_H(z)^2\left[\dfrac{1}{2^2} - \dfrac{1}{3^2}\right] \Rightarrow$ Ist line of Balmer series

$\bar{V}_2 = R_H(z)^2\left[\dfrac{1}{4^2} - \dfrac{1}{5^2}\right] \Rightarrow$ Ist line of Brackett series

$\dfrac{\bar{V}_1}{\bar{V}_2} = \dfrac{500}{81} \Rightarrow \boxed{5:0.81}$

53. Which of the following is correct set of 4 quantum number of 19th electron in Chromium (Atomic number = 24) in accordance with Aufbau principle?

(1) $n = 3, \ell = 2, m = +2, s = +\dfrac{1}{2}$

(2) $n = 3, \ell = 2, m = -2, s = +\dfrac{1}{2}$

(3) $n = 4, \ell = 1, m = 0, s = +\dfrac{1}{2}$

(4) $n = 4, \ell = 0, m = 0, s = +\dfrac{1}{2}$

Sol. Cr $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

$\Rightarrow$ 19th e^- belongs to 4s subshell

$n = 4 ; \ell = 0 ; m = 0 ; m_s = +\dfrac{1}{2} \text{ or } -\dfrac{1}{2}$

54. Given below are two statements:

Statement I : For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at constant pressure.

Statement II : In a constant volume process, no work is produced and all the heat withdrawn goes into the chaotic motion and is reflected by a temperature increase of the ideal gas.

In the light of the above statements, choose the ***correct*** answer from the options given below

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

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Sol. From first law of thermodynamics

$\Delta U = q + w$

constant volume $w = 0$

$\Delta U = q$

Also $C_P = C_V + R$

C_P = heat capacity at constant pressure

C_V = Heat capacity at constant volume.

R = constant

55. At T(K), the equilibrium constant of

$A_2(g) + B_2(g) \rightleftharpoons C(g)$ is 2.7×10^{-5}.

What is the equilibrium constant for

$\frac{1}{3}A_2(g) + \frac{1}{3}B_2(g) \rightleftharpoons \frac{1}{3}C(g)$ at the same temperature?

(1) $(2.7 \times 10^{-5})^3$ (2) 6×10^{-2}

(3) $\sqrt{2.7 \times 10^{-5}}$ (4) 3×10^{-2}

Sol. $A_2(g) + B_2(g) \rightleftharpoons (g)$ $K_{eq} = 2.7 \times 10^{-5}$

$\frac{1}{3}A_2(g) + \frac{1}{3}B_e(g) \rightleftharpoons \frac{1}{3}C(g) K^1_{eq} = \left(K_{eq}\right)^{1/3}$

$K^1_{eq} = \left(2.7 \times 10^{-5}\right)^{1/3}$

$= 3 \times 10^{-2}$

56. In order to oxidise a mixture of 1 mole each of FeC_2O_4, $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is :

(1) 3 (2) 2

(3) 5 (4) 7

Sol. Meq of KM_nO_4 = meq of FeC_2O_4 + meq of $Fe_2(C_2O_4)$ + meq of $FeSO_4$

moles $\times 5 = 1 \times 3 + 1 \times 6 + 1 \times 1$

moles $= \frac{10}{5} = 2$

57. Consider the first order reaction R $\rightarrow$ P.

The fraction of molecules decomposed in the given first order reaction can be expressed as

(1) $1 - e^{k_1 t}$ (2) $1 + e^{k_1 t}$

(3) $1 + e^{-k_1 t}$ (4) $1 - e^{-k_1 t}$

Sol. Ist order of Rμ^H

$a - x = ae^{-K_1 t}$

$\frac{x}{a} = 1 - e^{-K_1 t}$

58. A monoatomic anion (A^-) has 45 neutrons and 36 electrons. Atomic mass, group in the periodic table and physical state at room temperature of the element (A) respectively are :

(1) 80, 17, liquid (2) 81, 16, solid

(3) 80, 16, gas (4) 81, 15, gas

Sol. X = Br ; Atomic mass = 80

Period 4

Br_2 exist as liquid.

59. Given below are two statements :

Statement I : The covalency of oxygen is generally two but it can exceed upto four. The oxidation state of oxygen in SO_2 is –2 and in OF_2 it is +2.

Statement II : The anomalous behaviour of oxygen when compared to the other elements of group 16 is due to its small size and high electronegativity .

In the light of the above statements, choose the ***correct*** answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

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Sol. Oxygen can have covalency 4

In SO_2 ; O.S. of O = –2

In OF_2 ; O.S. of O = +2

Anamolous behaviour of oxygen is due to high EN and small size.

60. The **correct** statements among the following are,

A. Mo(VI) and W(VI) are less stable than Cr(VI).

B. Ce^{4+} and Tb^{4+} are oxidant while Eu^{2+} and Yb^{2+} are reductant.

C. Cm and Am have seven unpaired electrons.

D. Actinoid contraction is greater from element to element than lanthanoid contraction.

Choose the correct answer from the options given below :

(1) A and B Only (2) C and D Only

(3) B and D Only (4) A and C Only

Sol. (A) Mo(VI) & W(VI) more stable than Cr(VI)

(B) In lanthanoids +3 oxidation state is more stable hance +4 ions act as oxidising agents and +2 ions act as reducing agents.

(C) Cm – [Rn] $5f^7\ 6d^1 7S^2$ = 8 unpaired electrons

Am – [Rn] $5f^7\ 6S^2$ = 7 unpaired electrons

(D) Actinoid contraction is more than lanthanoid contraction

61. **Correct** statements from the following are

A. Potassium dichromate is an oxidising agent and it oxidises $FeSO_4$ to $Fe_2(SO_4)_3$ in acidic medium.

B. Sodium dichromate can be used as primary standard in volumetric estimation.

C. CrO_4^{2-} and $Cr_2O_7^{2-}$ are interconvertible in aqueous solution by varying the pH of the solution.

D. Cr–O–Cr bond angle in $Cr_2O_7^{2-}$ is 126°.

Choose the correct answer from the options given below :

(1) A, B and C only (2) A, C and D only

(3) A and C only (4) B and D only

Sol. **(A)** $MnO_4^- + Fe^{+2} + H^+ \longrightarrow Fe^{+3} + Mn^{+2} + H_2O$

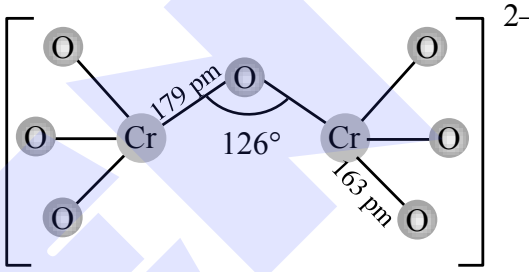
(B) $Na_2Cr_2O_7$ generally not used as a primary standard in volumetric analysis because it is deliquescent.

(C) In acidic medium $Cr_2O_7^{2-}$ exists and it get converted into CrO_4^{2-} in basic medium.

$$2CrO_4^{2-} + 2H^+ \longrightarrow Cr_2O_7^{2-} + H_2O$$

$$Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$$

(D) $K_2Cr_2O_7$ is a good oxidising agent and bond angle of Cr–O–Cr bond in $Cr_2O_7^{2-}$ is 126°



Dichromate ion

62. Match the **List-I** with **List-II**

List-I Complex ion		List-II Calculated spin only magnetic moment (BM)	
A.	$[Cr(H_2O)_6]^{2+}$	I.	3.87
B.	$[Co(H_2O)_6]^{2+}$	II.	5.92
C.	$[Cu(H_2O)_6]^{2+}$	III.	4.90
D.	$[Mn(H_2O)_6]^{2+}$	IV.	1.73

Choose the ***correct*** answer from the options given below:

(1) A-I, B-III, C-IV, D-II

(2) A-II, B-I, C-III, D-IV

(3) A-IV, B-II, C-I, D-III

(4) A-III, B-I, C-IV, D-II

Sol. $[Cr(H_2O)_6]^{2+} \rightarrow$ 4 unpaired e^- ; 4.90 BM(III)

$[Co(H_2O)_6]^{2+} \rightarrow$ 0 unpaired e^- ; 3.87 BM(I)

$[Cu(H_2O)_6]^{2+} \rightarrow$ 1 unpaired e^- ; 1.73 BM(IV)

$[Mn(H_2O)_6]^{2+} \rightarrow$ 5 unpaired e^- ; 5.92 BM(II)

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63. Increasing order of electron withdrawing power of following functional groups is :

a. –CN

b. –COOH

c. $-NO_2$

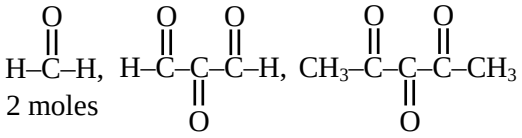
d. –I

(1) c < b < d < a (2) c < a < b < d

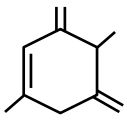
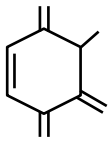
(3) d < b < a < c (4) a < b < c < d

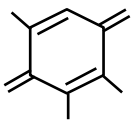
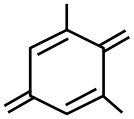
Sol. Theory based

64. An alkene (X) on ozonolysis followed by reduction gives following products.

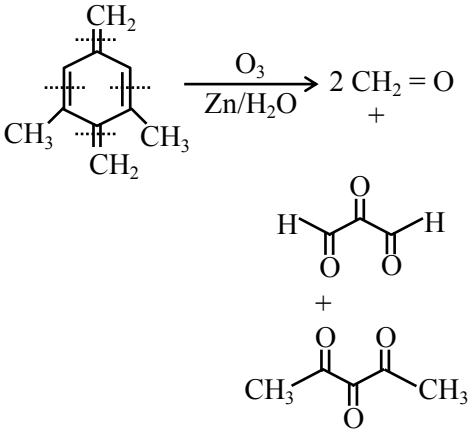


The alkene (X) is :

(1)  (2) 

(3)  (4) 

Sol.



65. Match the **List-I** with **List-II**

List_I Name of reaction		List-II Reagent or catalyst used	
A.	Finkelstein reaction	I.	SbF_3
B.	Swarts reaction	II.	Na, dry ether
C.	Sandmeyer's reaction	III.	NaI
D.	Fitting reaction	IV.	Cu_2Cl_2

Choose the ***correct*** answer from the options given below :

(1) A-I, B-IV, C-III, D-III

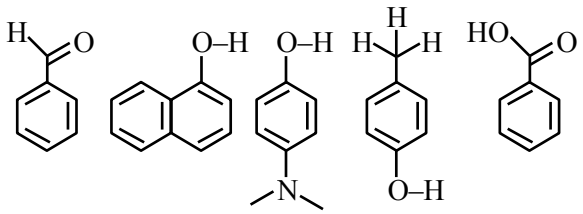
(2) A-III, B-I, C-IV, D-II

(3) A-IV, B-II, C-I, D-III

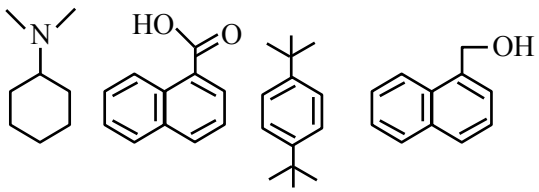
(4) A-I, B-III, C-II D-IV

Sol. Theory based

66. Amongst the following, the total number of compounds soluble in aqueous NaOH at room temperature is:



(I) (II) (III) (IV) (V)



(VI) (VII) (VIII) (IX)

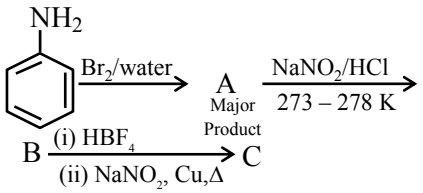
(1) 5 (2) 4 (3) 6 (4) 3

Sol. II, III, IV, V, VII

Compounds which are more acidic than H_2O can soluble in NaOH.

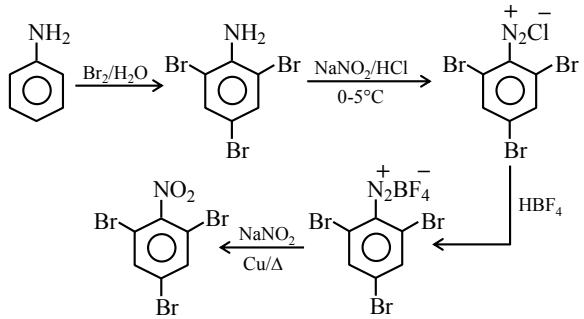
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67. Product C of the following reaction sequence will be



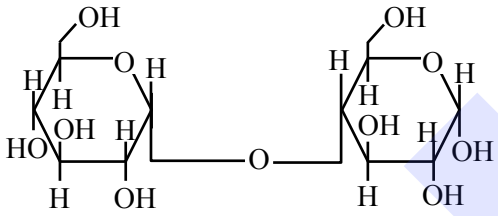
(1) 1-Bromo-4-nitrobenzene
(2) 1,3,5-Tribromo-2-nitrobenzene
(3) 4-Bromo-1-nitrobenzene
(4) 1,3,5-Tribromobenzene

Sol.



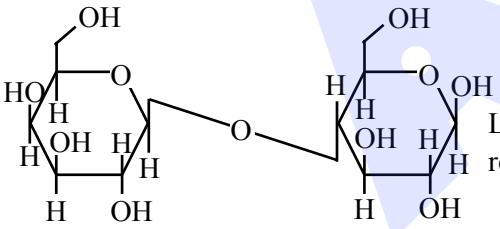
1,3,5-Tribromo
-2-nitrobenzene

68. **Statement-I :** The structure of Maltose is given below :



Maltose is a non reducing sugar

Statement-II : The structure of Lactose is given below



Lactose is a reducing sugar

In the light of the above statements, choose the ***correct*** answer from the options given below
(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

Sol. Maltose and lactose both have hemiacetal linkage. So both are reducing sugars.

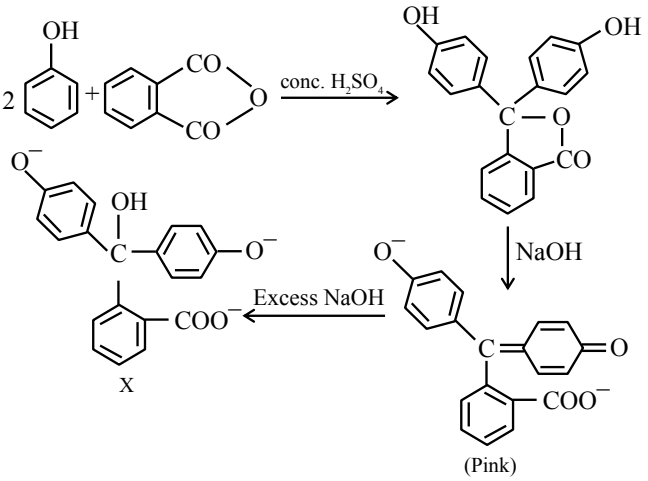
69. Match the **LIST-I** with **LIST-II** :

	List-I Name of amino acid		**List-II One letter symbol/type)**
A	Arginine	I.	D / Non-essential
B	Aspartic acid	II.	R / Essential
C	Lysine	III.	E / Non-essential
D	Glutamic acid	IV.	K/Essential

Choose the ***correct*** answer from the options given below :
(1) A-II, B-I, C-IV, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-IV, C-I, D-III

Sol. Arginine (R) and Lysine (K) are essential amino acid. Aspartic acid (D) and Glutamic acid (E) are non-essential amino acid.

70. Identify the colour of compound 'X' in the sequence of the reaction.



(1) Violet
(2) Green
(3) Red
(4) Colourless

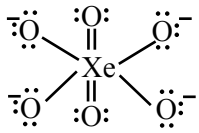
Sol. Phenolphthalein indicator turns colourless in excess NaOH.

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SECTION-B

71. According to Lewis theory, the total number of σ bond-pairs and lone pair of electrons around the central atom of XeO_6^{4-} ion is ____.

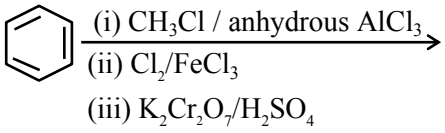
Sol.



σ = 6

Lone pairs = 0

72. Consider the following sequence of reactions to give the major product (X)

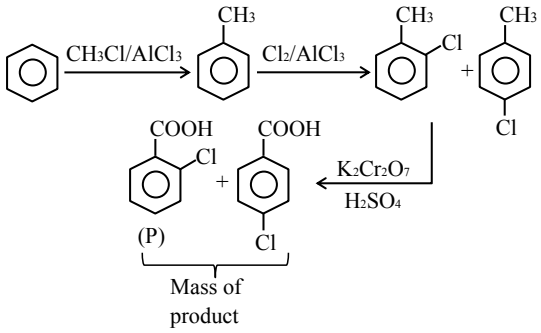


P g of the major product (X) formed is reacted with $NaHCO_3$soluion to liberate a gas which occupied 11.2 dm^3 at STP.

P = ____. g.

(Given molar mass in g mol^{-1} H : 1, C : 12, O : 16, Cl : 35.5)

Sol.



$$\text{Moles of P} = \frac{11.2}{22.4} = 0.5 \text{ mole}$$

$$M_{C_7H_5O_2Cl} = 156.5 \text{ gm}$$

$$W_{P(C_7H_5O_2Cl)} = \frac{156.5}{2} = 78.25 \approx 78 \text{ grams}$$

73. 2.0 g of a bromo hydrocarbon (X) was subjected to Carius analysis, gave 3.36 g of AgBr. The percentage of carbon in the compound (X) is 26.7%. Total number of carbon atoms in the empirical formula for compound (X) is ____.

(Given molar mass in g mol^{-1} H : 1, C : 12, Br : 80, Ag : 108)

Sol. $n_{Br} = \frac{3.36}{(108+80)}$

r = no. of Br atoms in organic compound

$$n_{\text{organic compound}} = \frac{3.36}{(108+80)\times x} = \frac{2}{M}$$

$$M = \frac{2\times 188\times x}{3.36}$$

M = ½ × x

(x is integer)

$$W_C = \frac{112\times x\times 26.7}{100} = 30 \times x$$

$$n_C = \frac{30\times x}{12} = 2.5 \times x$$

take x = 2, $n_C = 5$

(Empirical formula is C_5H_4Br)

74. The pH a solution obtained by mixing 5 mL of 0.1 M NH_4OH solution with 250 mL of 0.1 M NH_4Cl solution is _____ × 10^{-2}. (Nearest integer)

Given : pK_b (NH_4OH) = 4.74

log 2 = 0.30

log 3 = 0.48

log 5 = 0.70

Sol. On mixing final volume = 255 ml

$$[NH_4OH] = \frac{5\times 0.1}{255}, [NH_4Cl] = \frac{250\times 0.1}{255}$$

$$p^{OH} = pk_b + \log\frac{[NH_4Cl]}{NH_4OH}$$

$$p^{OH} = 4.74 + \log\frac{250\times 0.1}{5\times 0.1}$$

$$P^H = 14 - P^{OH} = 14 - 6.44$$

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75. A non-volatile, non-electrolyte solid solute when dissolved in 40 g of a solvent, the vapour pressure of the solvent decreased from 760 mm Hg to 750 mm Hg. If the same solution boils at 320 K, then the number of moles of the solvent present in the solution is _____. (Nearest integer)

[Given : boiling point of the pure solvent = 319.5 K, K_b of the solvent = 0.3 K kg mol^{-1}]

Sol. $$\frac{P^\circ - Ps}{Ps} = i.\ \text{molality} \times \frac{(M.\text{solvent})}{1000}$$

$$\Delta T_b = i.\ K_b.\ \text{molality} \Rightarrow \text{molality} = \frac{0.5}{0.3}$$

$$(\text{Molecular Mass}) = \frac{600}{75}\text{g}$$

$$\text{Moles} = \frac{40}{600/75} = 5$$

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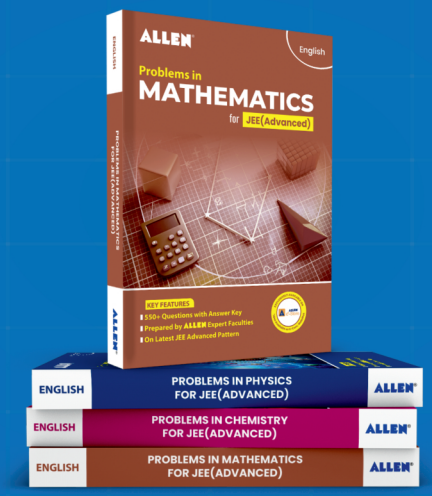


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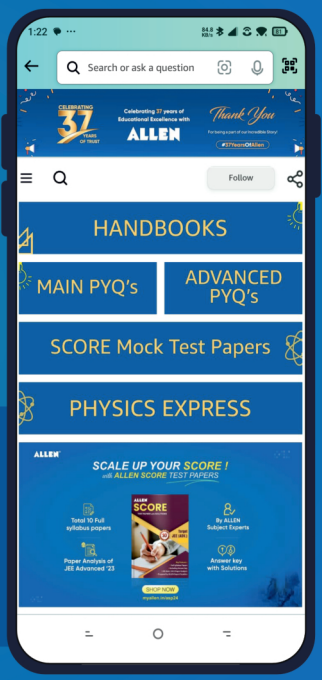
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